

Multi-Dimensional Cross-Granularity Open-Set Network Intrusion Detection

Minh-Thuyen Thi^{ID} and Mohan Gurusamy^{ID}, *Senior Member, IEEE*

Abstract—Network intrusion detection systems (NIDSs) face critical challenges from continuously evolving cyber-attacks. Traditional machine learning methods, while requiring extensive labeled training data, still often fail against unknown and out-of-distribution (OOD) attacks. Furthermore, new sophisticated adversaries are exploiting the detection blind spots inherent in traditional feature representation approaches that do not provide adequate comprehensive traffic analysis. In this paper, we propose MDCG-IDS, an NIDS framework that introduces multi-dimensional cross-granularity (MDCG) feature representation for open-set detection, in which network traffic is analyzed thoroughly across three complementary dimensions (traffic statistics, temporal, spatial), each at multiple granularity levels. These dimensions and granularities jointly capture the structures of sophisticated attacks that may be invisible from single analytical perspectives. We design a tensor structure that provides a unified encoding for the MDCG features while supporting the use of optimal transport theory to measure the distance between benign traffic and known or unknown attacks. MDCG-IDS uses a semi-supervised learning model that is trained exclusively on benign traffic and validated on a small set of labeled data, significantly reducing the effort of data labeling. Experiments on various datasets achieve AUC-ROC scores of more than 0.948, exceeding the best competing state-of-the-art methods by up to 7%. Regarding the amount of labeled validating data, MDCG-IDS obtains an AUC-ROC score of over 0.94 with only 3% of entire validating samples, outperforming the baseline models.

Index Terms—Network intrusion detection, open-set intrusion detection, semi-supervised learning, optimal transport, multi-granularity analysis.

I. INTRODUCTION

THE continuous evolution of cyber-attacks has made traditional network intrusion detection systems (NIDSs) inadequate for contemporary security challenges [1], [2]. Every second, three to four new malware variants are discovered [3], creating an inherent open-set problem in which possible attacks remain unbounded and are constantly expanding. Conventional supervised NIDS approaches suffer from critical limitations. First, they require extensive labeled datasets containing representative samples from all attack

categories, which creates an expensive and time-consuming data collection [4]. Moreover, network security data involve privacy constraints and proprietary concerns, leading to limited dataset diversity [5]. Attack data also remain incomplete, as they inevitably contain noise from misclassified activities and adversarial evasion techniques [6]. Second, even with labeled datasets, the closed-set classification paradigm often fails when faced with unknown attacks (i.e., new attack types) and out-of-distribution (OOD) class categories (i.e., categories that were not covered by the distribution of the training data) [7], [8]. This is particularly problematic when zero-day attacks represent critical threats without prior training examples. Finally, since adversaries can bypass the NIDSs that use traditional feature representations [6], [9], we need a comprehensive feature analysis approach to capture more diverse aspects of network traffic. Such an approach must effectively learn the network traffic distribution to precisely measure the distances between benign samples and known or unknown attacks.

To deal with the problem of limited labeled data, this paper introduces MDCG-IDS, a distance-based anomaly detection framework within an open-set setting. Unlike supervised approaches that require comprehensive labeled attack datasets, our semi-supervised methodology builds reference models from benign traffic samples—similar to one-class classification but enhanced with a small labeled validating set for hyperparameter tuning. This approach aligns with binary detection in open-set NIDS, where the objective is to distinguish between benign and unknown/OOD malicious traffic [10].

To address the requirements for an efficient feature representation in a data-limited open-set NIDS, we propose a feature analysis approach that takes into account multiple analytical dimensions and granularities. On the one hand, multi-dimensional methods have demonstrated their effectiveness in capturing the complex inter-dependencies of network traffic [11], [12], [13], [14] (i.e., spatial and temporal dimensions). This is because single-dimensional methods may miss attacks that appear benign from one perspective while exhibiting anomalous behavior from another. For example, an Advanced Persistent Threat (APT) may appear benign at the packet level while exhibiting suspicious cross-flow patterns at the session level and anomalous protocol sequences in the temporal dimension. On the other hand, multi-granularity feature analysis has shown its advantages over single-granularity approaches in [15] (multi-instance learning that combines bag-level and instance-level granularities), [16] (host-level intrusion detection that combines system call-level and behavior-level granularities), and [17]

Received 13 November 2025; revised 19 March 2026 and 30 April 2026; accepted 8 May 2026. Date of publication 13 May 2026; date of current version 19 May 2026. This research is supported by A*STAR, CISCO Systems (USA) Pte. Ltd and the National University of Singapore under its Cisco-NUS Accelerated Digital Economy Corporate Laboratory (Award I21001E0002). The associate editor coordinating the review of this article and approving it for publication was K. Hammar. (*Corresponding author: Minh-Thuyen Thi.*)

The authors are with the Department of Electrical and Computer Engineering, National University of Singapore, Singapore 119077 (e-mail: tmthuyen@gmail.com).

Digital Object Identifier 10.1109/TNSM.2026.3693141

(auto-encoder that can learn multiple granularities of feature abstraction). As a result, we develop a multi-dimensional cross-granularity (MDCG) architecture which examines both benign and known/unknown attacks through three complementary perspectives: (i) the traffic dimension, which captures numerical patterns across packet, flow, and session-levels; (ii) the temporal dimension, which analyzes symbolic feature sequences across multiple time-scales; and (iii) the spatial dimension, which examines symbolic attributes across session and host-level granularities. This comprehensive decomposition enables the analysis of attacks on different statistical, temporal, and spatial scales. To encode these multi-perspective features, we introduce a unified MDCG tensor structure that enables vectorized computations across granularity levels simultaneously. This representation preserves the relationships across dimensions and simultaneously supports efficient batch processing and parallel computation.

We use *optimal transport* [18] to quantify the distributional deviations between attack samples and benign traffic. Given that there exist several studies employing optimal transport for detecting outlier/anomalous data in other research fields (e.g., image processing, audio processing), MDCG-IDS is, to the best of our knowledge, the first application of optimal transport in network intrusion detection. This framework allows us to compare probability distributions that capture both the magnitude and geometric structure of the multi-dimensional shifts between traffic samples. For example, when benign traffic and an attack have similar average packet sizes, traditional statistical measures (mean, variance) might consider them equivalent, whereas optimal transport can reveal the difference in how the probability mass is distributed across packet sizes. The design of MDCG tensor structure also allows optimal transport to quantify such difference for each individual granularity level, then combine features across dimensions and granularities. Distance measurement is performed using the Wasserstein metric through Sinkhorn-Knopp algorithms [19] to reduce complexity and enable real-time intrusion detection. This framework accommodates both numerical and symbolic features through specialized formulations: sliced Wasserstein distances for numerical features and Gromov-Wasserstein distances for symbolic features.

The main contributions of this paper are summarized as follows:

- We model a novel multi-dimensional cross-granularity framework that decomposes network traffic into three complementary analytical perspectives and multiple granularity levels, enabling comprehensive analysis and comparison between benign and known/unknown/OOD attacks. These dimensions are unified through tensor representations that support vectorized computations across heterogeneous feature types.
- To the authors' knowledge, this study is the first application of optimal transport theory to network intrusion detection, providing a mathematical foundation for cross-granularity anomaly scoring through Wasserstein distances. We define specialized formulations of sliced Wasserstein distances for numerical features and Gromov-Wasserstein distances for symbolic features. The Sinkhorn-Knopp approximations are proposed to

facilitate real-time processing suitable for production network environments.

- We design a practical framework that achieves strong OOD detection performance while being trained exclusively on benign traffic with minimal labeled validating data. Extensive evaluations demonstrate effective cross-dataset performance, robustness against data contamination, and real-time processing capability. These results further validate the framework's deployability in operational environments.

The remainder of the paper is organized as follows. Section II discusses related works; Section III presents MDCG feature representation; Section IV describes the optimal transport-based OOD detection methodology; Section V provides experimental evaluations; Section VI presents discussions and future works; finally, Section VII concludes with future directions.

II. RELATED WORK

This section presents related work across three key areas: closed-set, open-set, and multi-granular intrusion detection.

A. Closed-Set Intrusion Detection

Early intrusion detection systems relied on rule-based signatures and statistical anomaly detection, while modern approaches leverage deep learning architectures based on popular frameworks such as Convolutional Neural Networks (CNN) [20], Recurrent Neural Networks (RNN) [21], [22], Graph Neural Networks (GNN) [23], [24], [25], autoencoders [26], and Generative Adversarial Networks (GAN) [27]. The study [22] proposed to combine attention mechanisms, Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU) for early flow classification in NIDSs (LS-GRU), leveraging the sequential nature of packets and allowing prompt attack detection without requiring flow termination. The authors in [24] proposed TCG-IDS, a self-supervised learning model designed based on a temporal contrastive GNN. This model integrates temporal, asymmetric, and masked contrasting strategies to learn the network by masking some data during training. Wei et al. [26] proposed AHDM, an autoencoder-based hybrid detection model with a dual-classifier scheme. In AHDM, autoencoder-enhanced features target small-sample malicious traffic, while a complementary classifier on raw features addresses normal and large-sample traffic. This methodology translates into superior performance on standard benchmarks, notably improving small-sample and DDoS detection over advanced baselines (e.g., Deep Neural Networks). These closed-set approaches do not provide the capability to reject new attack categories that are not seen during training. Instead, they force classification into predefined categories, even for novel attacks.

B. Open-Set Intrusion Detection

Traditional approaches for open-set NIDS include One-class Support Vector Machine (OSVM) [28] and Isolation Forest [29]. Recent deep learning advances have introduced new methodologies based on architectures such as Isolation Trees [30], CNN [31], auto-encoders/variational auto-encoders [32], [33], and extreme value theory [32], [33]. The study in

[30] introduced FOSS, a fine-grained unknown class detection model against variable attack spectra, incorporating adaptive contamination modeling. FOSS leverages an isolation-tree backbone with hyperplane splitting, isolation-based discrimination, and incremental updates to obtain accurate fine-grained detection of previously unseen DDoS attacks. In [31], the authors propose a scalable IDS for detecting unknown attacks by combining a CNN with a nearest class mean (NCM) classifier, jointly optimized by Fisher and maximum mean discrepancy losses. They further introduce semantic-embedding clustering to reveal latent unknown classes and an incremental nearest cluster centroid method to assimilate them by updating the NCM classifier. The study [32] formulates fine-grained known/unknown intrusion detection as a two-stage problem using class-conditioned autoencoders. They model reconstruction-error tails via extreme value theory and benign clustering to boost unknown-attack detection. Other open-set approaches include MinMax auto-encoder with pseudo extreme value [33], deep reinforcement learning with self-evolving moving targets [34], Mahalanobis distancing [35], Dandelion Network [36], and image-based representations of packet-level data [37].

C. Multi-Dimensional and Cross-Granularity Feature Analysis

Recent research has leveraged multi-dimensional models for intrusion detection by applying spatio-temporal feature analysis. Wang et al. [11] developed an attention-weighted spatial-temporal graph model to extract features in parallel using a message-passing mechanism and a Bi-LSTM, then fuse them for the final classification of intrusions. The authors showed that a comprehensive spatial-temporal correlation approach helps to compensate when single-dimensional extraction yields weak representations. The authors in [12] proposed an attention-based Spatial Temporal Graph Convolutional Network (ST-GCN) to model normal link states in networks. Extreme Value Theory (EVT) is used to identify Link Flooding Attacks (LFA) as extreme performance degradation events. The authors demonstrated that modeling network states as a spatio-temporal graph is more efficient than modeling with solely temporal (LSTM) or spatial (GCN) dimension. The study [13] introduced an anomaly detection framework for 5G that captures anomalous patterns across multiple spatio-temporal perspectives using a Multi-Scale Fuzzy Contrastive Learning approach. It performs anomaly detection explicitly in the latent space by measuring the distance to normal data prototypes. Yang et al. [14] used spatio-temporal context-aware graph embedding to learn representations of normal behavior, from which an anomaly score is generated for outlier detection. Similar to [11], the authors in [13] and [14] showed that spatio-temporal approach allows to extract more comprehensive traffic patterns; especially when combining with multi-scale feature analysis [13].

Several studies have explored multi-granularity approaches to capture network behavior with different feature scales. Geng et al. [17] proposed S3WD-MG, a multi-granularity intrusion detection algorithm based on sequential three-way decisions. This algorithm incorporates autoencoder-based feature learning across different granularity levels. The authors in [16] developed BR-HIDF, an anti-sparsity host intrusion

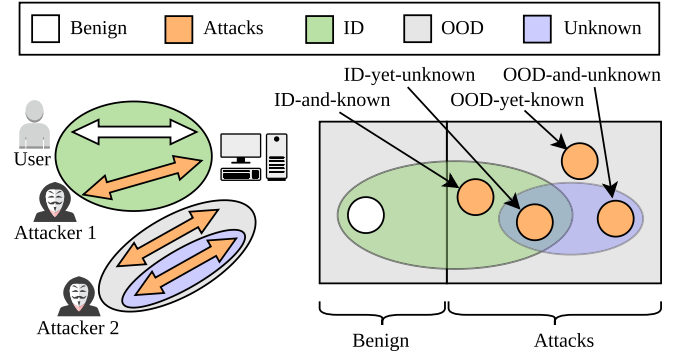


Fig. 1. Open-set scenario showing OOD and unknown traffics from Attacker 2, and in-distribution (ID) traffic from Attacker 1. The diagram on the right depicts a conceptual illustration of attack categories, in which the *ID-yet-unknown* attacks are the samples that coincidentally have similar structures with attacks that have been learned; while the *OOD-yet-known* attacks are the threats that have been known in the cybersecurity literature but were not covered by the training dataset due to limited coverage.

detection framework that employs multi-granularity feature extraction to address the challenges of sparsity in host-based detection. Jiang et al. [15] introduced multi-granularity abnormal traffic detection based on multi-instance learning, capturing traffic patterns at various temporal resolutions. In these studies, the multi-granularity approach helps to avoid premature decisions under insufficient information [17], and allows multi-instance learning to flexibly analyze the traffic at either bag-level or instance-level, providing adaptive anomaly detection [15]. However, existing multi-granularity approaches primarily focus on the temporal aspects of the closed-set problem, and lack comprehensive multi-dimensional analysis across multiple perspectives simultaneously. Unlike our framework, those approaches do not utilize a multi-dimensional cross-granularity representation to model the open-set NIDS and support the quantification of OOD levels.

D. Summary and Comparisons

Current approaches suffer from several limitations: (1) both multi-dimensional and cross-granularity approaches are applied only to closed-set detection; hence, open-set methods lack comprehensive multi-dimensional cross-granular representations that integrate traffic, temporal, and spatial perspectives simultaneously, and (2) traditional distance metrics typically rely on single-dimensional deviation without capturing multi-dimensional geometric relationships between traffic samples. In this paper, we measure that relationship by designing a multi-dimensional distance metric based on optimal transport theory.

Table I compares MDCG-IDS with existing methods across key criteria. Our MDCG framework uniquely addresses the criteria of open-set recognition capacity, multi-dimensional cross-granularity analysis, and optimal transport-based distance computation.

III. MULTI-DIMENSIONAL CROSS-GRANULARITY (MDCG) FEATURE REPRESENTATION

A. OOD and Unknown Attacks Detection

The open-set intrusion detection of OOD and unknown attacks is illustrated in Fig. 1. The figure depicts an example of

TABLE I
SUMMARY OF NETWORK INTRUSION DETECTION APPROACHES ON OPEN-SET DETECTION CAPABILITY, FEATURE REPRESENTATION, DISCRIMINATION METHOD, AND LEARNING PARADIGM

Method	Open-Set	Multi-Dimension	Cross-Granularity	Discrimination Method	Learning Paradigm
TCG-IDS [24]	×	×	×	Contrastive	Self-supervised, supervised
LS-GRU [22]	×	×	×	×	Supervised
AHDM [26]	×	×	×	×	Unsupervised, supervised
Multi-dimensional methods [11]	×	Spatio-temporal	×	×	Supervised
Multi-dimensional methods [12]	×	Spatio-temporal	×	Extreme Value Theory	Unsupervised
Multi-dimensional methods [13]	×	Spatio-temporal	×	Euclidean, Hamming, cosine	Self-supervised
Multi-dimensional methods [14]	×	Spatio-temporal	×	Graph-based score	Supervised
Cross-granularity methods [15]–[17]	×	×	✓	×	Supervised
OIDMD [35]	✓	×	×	Mahalanobis	Supervised
OSDN [36]	✓	×	×	Cosine	Unsupervised
OpenIDS [33]	✓	×	×	Reconstruction loss	Unsupervised
FOSS [30]	✓	×	×	Isolation-based distinction	Semi-supervised
MDCG-IDS	✓	✓	✓	Optimal transport	Semi-supervised

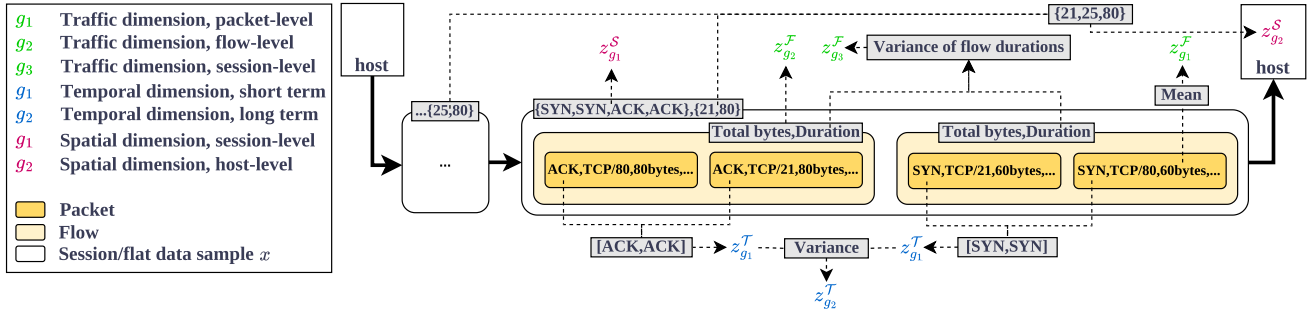


Fig. 2. Illustration of the dimension-specific encoding, showing data extraction to construct MDCG features. The illustration shows an example of two network sessions between two hosts, one of the sessions includes two network flows, each has two packets. The network packets have attributes of TCP flag, transport protocol, destination port, and packet size. Only a selected list of feature examples in Table III are illustrated.

benign traffic from a legitimate user and attack traffic from two attackers: Attacker 1 generates in-distribution (ID) samples, while Attacker 2 creates OOD and unknown samples. The diagram on the right provides a conceptual illustration of four attack categories: *ID-and-known*, *ID-yet-unknown*, *OOD-yet-known*, and *OOD-and-unknown*. ID attacks share structural similarities with learned patterns, whereas OOD attacks fall outside the training coverage, with some being completely novel (unknown).

B. Feature Granularity Definitions

We define the granularity levels for traffic, temporal, and spatial dimensions as $\mathcal{G}^{\mathcal{F}} = \{g_1^{\mathcal{F}}, g_2^{\mathcal{F}}, g_3^{\mathcal{F}}\}$, $\mathcal{G}^{\mathcal{T}} = \{g_1^{\mathcal{T}}, g_2^{\mathcal{T}}, \dots\}$, and $\mathcal{G}^{\mathcal{S}} = \{g_1^{\mathcal{S}}, g_2^{\mathcal{S}}\}$, respectively. The summary of the main mathematical symbols is given in Table II. The following sections describe in detail these dimensions and granularities, which are also illustrated in Fig. 2. The figure describes an example of the dimension-specific encoding of network traffic between two hosts. In the example, the flat features are extracted to construct MDCG features at different dimensions and granularities.

1) *Traffic Dimension*: The traffic dimension $\mathcal{F}$ focuses exclusively on numeric statistical features that characterize quantitative patterns across three granularity levels: packet, flow, and session. In Fig. 2, those features are mean packet size, flow duration, and inter-flow time variance. The three granularities are defined as:

- Packet-level ($g_1^{\mathcal{F}}$): extracts first-order statistical features from individual packets, such as mean packet size and

TABLE II
MAIN MATHEMATICAL NOTATIONS

Symbol	Definition
x_i	Flat data sample corresponding to network session i
z_i	MDCG training tensor of sample i
$\mathcal{Z}$	Benign training set $\{z_i, i = 1, \dots, n\}$
$\hat{z}$	Incoming sample (either validating or testing sample)
D	Number of dimensions, i.e., $D = 3$ for the set $\{\mathcal{F}, \mathcal{T}, \mathcal{S}\}$
$\mathcal{G}^{\mathcal{D}}$	Set of granularity levels for a dimension $\mathcal{D} \in \{\mathcal{F}, \mathcal{T}, \mathcal{S}\}$
$G_{\max}$	Maximum number of granularities across dimensions
$F_{\max}$	Maximum number of features across dimensions and granularities
g	A granularity level, $g \in \mathcal{G}^{\mathcal{D}} = \{g_1^{\mathcal{F}}, g_2^{\mathcal{F}}, g_3^{\mathcal{F}}\} \cup \{g_1^{\mathcal{T}}, g_2^{\mathcal{T}}\} \cup \{g_1^{\mathcal{S}}, g_2^{\mathcal{S}}\}$
$\Pi_g^{\mathcal{D}}$	Projection operator that extracts features with dimensionality $d_g^{\mathcal{D}}$ at the granularity level g of dimension $\mathcal{D}$
$\rho_g^{\mathcal{D}}(\cdot, \cdot)$	Distance function (either Sinkhorn-Wasserstein or Gromov-Wasserstein) at $\mathcal{D}$ and g
$\epsilon_g^{\mathcal{D}}$	Regularization parameter at $\mathcal{D}$ and g in the Sinkhorn-Wasserstein distance
$\alpha_g^{\mathcal{D}}$	Granularity-specific weight at $\mathcal{D}$ and g in the computation of OOD score
$\beta^{\mathcal{D}}$	Non-negative weight at $\mathcal{D}$ in the computation of final OOD score
Θ	Complete parameters set, including $\{\alpha_g^{\mathcal{D}}\}$ and $\{\beta^{\mathcal{D}}\}$
$\psi_{\text{final}}(\hat{z})$	Final OOD score of incoming sample $\hat{z}$

mean inter-arrival time. This granularity level helps to detect attacks that exhibit abnormal packet sizes, such as the Ping of Death (PoD).

- Flow-level ($g_2^{\mathcal{F}}$): aggregates packet statistics within network flows, creating features such as per-flow total bytes

TABLE III
SUMMARY OF MDCG FEATURES, SHOWING THEIR DIMENSION,
GRANULARITY, DATA TYPES AND EXAMPLES

Dimension	Granularity	Type	Example	Attack example
Traffic	Packet ($g_1^{\mathcal{F}}$)	Numeric	Mean packet size, mean inter-arrival time	PoD
	Flow ($g_2^{\mathcal{F}}$)	Numeric	Per-flow total bytes, flow duration	DoS
	Session ($g_3^{\mathcal{F}}$)	Numeric	Variance of flow duration, ratio of forward-backward packets	APTs
Temporal	Short ($g_1^{\mathcal{T}}$)	Symbolic	Sequence of TCP flags, sequence of transport protocols	Port scan
	Long ($g_2^{\mathcal{T}}$)	Symbolic	Variance of lower-level granularity TCP flag counts/transport protocol counts	APTs
Spatial	Session ($g_1^{\mathcal{S}}$)	Symbolic	Set of per-session TCP flags	Connection resets
	Host ($g_2^{\mathcal{S}}$)	Symbolic	Set of per-host destination ports, set of per-host application protocols	Service exploit

and flow duration. This granularity level targets attacks such as DoS, which have an unusual traffic volume.

- Session-level ($g_3^{\mathcal{F}}$): captures statistical relationships across multiple flows within sessions, for example, variance of flow durations and ratio of forward-backward packets. This granularity level aims to identify attacks such as APTs that exhibit suspicious cross-flow patterns.

2) *Temporal Dimension*: The temporal dimension $\mathcal{T}$ represents sequences of symbolic features across multiple configurable timescales, capturing procedural and sequential patterns. The MDCG framework defines flexible multi-level timescales $g_i^{\mathcal{T}} \in \mathcal{G}^{\mathcal{T}}$ to cover different temporal characteristics: short granularity captures sequential patterns within small time windows, such as symbolic sequence of TCP flags and sequence of transport protocols, while long granularity aggregates occurrences of those sequences in larger time windows, e.g., variance of lower-level granularity TCP flag counts/transport protocol counts (Fig. 2). This multi-scale approach provides coarse-grained granularities to identify persistent threats (APTs), as well as fine-grained granularities to detect fast-moving attacks, such as port scans. The exact number of timescales can be decided during deployment to suit different network environments.

3) *Spatial Dimension*: The spatial dimension $\mathcal{S}$ represents non-sequential sets of symbolic features that characterize the data exchange at the session and host-levels. These two granularities are detailed as follows. Session-level ($g_1^{\mathcal{S}}$) contains symbolic features that characterize individual sessions, such as set of per-session TCP flags (Fig. 2). This level facilitates the detection of attacks such as *abnormal connection resets*. Host-level ($g_2^{\mathcal{S}}$) aggregates session-level features into per-host attributes, such as set of per-host destination ports and set of per-host application protocols (Fig. 2). This level helps to detect compromised hosts that exhibit unusual communication

patterns, such as accessing unexpected application services (*service exploit*).

Table III summarizes the MDCG features, including their dimensions, granularities, data types, and examples.

C. MDCG Feature Tensor Structure

We introduce a new tensor structure to encode the MDCG features, preserving the relationships between different dimensions and granularity levels, while adapting to the computation of optimal transport-based distances. By using vectorized operations, the representation supports batch processing of distance computations across multiple granularities simultaneously. These tensors can be processed with GPU acceleration to improve scalability with respect to the number of dimensions and granularity.

1) *Unified Tensor Structure*: In this paper, each flat data sample $x_i, i = 1, 2, \dots$ (i.e., either training, validating, or testing sample) typically corresponds to a network session. We note that training samples include only benign traffic records, while validating samples are small quantity of labeled records, and testing samples are labeled records used for inference. For each sample, we define a unified tensor representation $z_i \in \mathbb{R}^{D \times G_{\max} \times F_{\max}}$ where D is the number of dimensions, $G_{\max} = \max\{|\mathcal{G}^{\mathcal{F}}|, |\mathcal{G}^{\mathcal{T}}|, |\mathcal{G}^{\mathcal{S}}|\}$ represents the maximum number of granularity levels across all dimensions, and $F_{\max}$ is the maximum number of features (Fig. 3). Each tensor element contains the features of a specific granularity level g within a dimension $\mathcal{D}$. This tensor is extracted through the dimension-specific encoding function as $z_i = \phi(x_i)$. Fig. 3 corresponds to the examples in Table III, in which $D = 3$ and $G_{\max} = 3$ with unused tensor positions are zero-padded to maintain consistency, and $F_{\max} = 2$ since no granularity in the examples of Table III has more than 2 features.

2) *Dimension-Specific Encoding Functions*: The encoding functions are designed to handle the heterogeneous features across different dimensions while ensuring compatibility with optimal transport computations. For the traffic dimension with exclusively numeric features, the function $\phi(x_i)$ applies normalization using the mean and standard deviation computed from the training data, followed by zero-padding to achieve uniform dimensionality $F_{\max}$. Before normalization, the granularity-specific numeric features are extracted from the flat data sample x_i using a projection operator.

Temporal and spatial dimensions contain symbolic features, with temporal features being sequential while spatial features are unordered. For the temporal dimension, we combine positional embeddings with learned categorical embeddings to maintain sequential relationships. The encoding function $\phi(\cdot)$ aggregates all embeddings at the granularity level g , preserving sequential order through weighted averaging based on temporal proximity. For the spatial dimension, we employ standard categorical embeddings without positional constraints, and the encoding function uses uniform averaging.

IV. MDCG OOD DETECTION

Our framework learns reference distributions from benign training data, along with a small amount of labeled validating traffic. During training, the MDCG anomaly scores are computed using optimal transport. Based on these scores, the

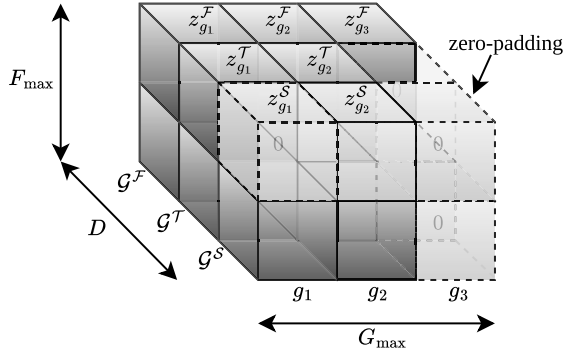


Fig. 3. Unified tensor structure corresponding to the examples in Table III. In those examples, $G_{\max} = 3$ (as there are 3 dimensions) and $F_{\max} = 2$ (as no granularity has more than 2 features). The tensor elements that have less than 3 dimensions or 2 features are zero-padded.

OOD detector estimates the Area Under the Curve of the Receiver Operating Characteristic (AUROC), and then performs parameter optimization using the labeled validating data by maximizing this AUROC. During inference, the detector applies the optimized parameters and thresholded scores for binary OOD detection.

A. MDCG-Decomposed Optimal Transport Distance

In this paper, we define an *incoming sample* as either a validating or testing sample. The distance metric quantifies the dissimilarity between an *incoming sample* $\hat{z}$ and the benign training distribution. We propose to use optimal transport (OT) [18] to leverage MDCG decomposition and Wasserstein distances. Optimal transport is a mathematical framework for measuring the minimum cost of transforming one probability distribution into another, where the cost reflects how much probability mass must be moved and how far. In the context of intrusion detection, OT quantifies how much the distribution of an incoming traffic sample differs from the benign reference distribution. The Wasserstein distance, derived from OT theory, provides a principled metric for this distributional comparison. Unlike statistical measures such as Euclidean distance, which use the mean and variance, OT and Wasserstein distance capture both the magnitude and geometric structure of distribution differences, aligning with MDCG traffic analysis. For instance, two traffic samples with similar mean packet sizes but different distributions are distinguished by how their probability masses are spatially arranged. In the following, we present the OT distance computation that is decomposed into different dimensions.

The Wasserstein distance is extended to operate across multiple granularity levels simultaneously. Let $\mathcal{Z} = \{z_i, i = 1, 2, \dots, n\}$ represent the benign samples of the training dataset, where z_i corresponds to a feature tensor. For each dimension $\mathcal{D} \in \{\mathcal{F}, \mathcal{T}, \mathcal{S}\}$ and granularity level $g \in \mathcal{G}^{\mathcal{D}}$, we define the projection operator $\Pi_g^{\mathcal{D}} : \mathbb{R}^{D \times G_{\max} \times F_{\max}} \rightarrow \mathbb{R}^{d_g^{\mathcal{D}}}$ that extracts features with dimensionality $d_g^{\mathcal{D}}$ at the granularity level g of dimension $\mathcal{D}$. For the training set $\mathcal{Z}$, its projections are denoted as $\mathcal{Z}_g^{\mathcal{D}} = \{z_{i,g}^{\mathcal{D}}, i = 1, \dots, n\}$, where $z_{i,g}^{\mathcal{D}} = \Pi_g^{\mathcal{D}}(z_i)$. Similarly, for an incoming sample $\hat{z}$, its projections are $\hat{z}_g^{\mathcal{D}} = \Pi_g^{\mathcal{D}}(\hat{z})$.

To handle the discrete features, we incorporate regularization into the optimal transport formulation. The regularization and distance computation are adapted with respect to dimension: the traffic dimension (i.e., numeric features) uses the Sinkhorn-Wasserstein (SW) distance, while the temporal and spatial dimensions (i.e., symbolic features) employ the Gromov-Wasserstein (GW) distance. This is because the formulation of SW distance requires direct cost (distance) between points, as shown in below formulas; hence it is suitable for numeric traffic features where pairwise distances are directly quantifiable. This distance measures how much numeric mass must be shifted to match two feature distributions. In contrast, the formulation of GW distance relies on the internal relational structures of two distributions rather than requiring direct point-to-point comparison. As a result, to measure the distance between two symbolic values (e.g., HTTP and FTP protocols), we use GW distance to compare how these categories relate to each other through their co-occurrence patterns. Both distances are computed using iterative algorithms: Sinkhorn-Knopp for SW, and alternating minimization for GW. The matrix-scaling algorithm Sinkhorn-Knopp can provide closed-form iterative updates for SW, while GW's quadratic structure requires alternating minimization for iterative optimization.

For an incoming sample $\hat{z}$, the Sinkhorn-Wasserstein distance $\rho_{\text{SW},g}^{\mathcal{F}}(\hat{z}_g^{\mathcal{F}}, \mathcal{Z}_g^{\mathcal{F}})$ corresponding to traffic dimension $\mathcal{F}$ is computed as:

$$\rho_{\text{SW},g}^{\mathcal{F}}(\hat{z}_g^{\mathcal{F}}, \mathcal{Z}_g^{\mathcal{F}}) = \min_{\gamma \in \Delta_n} \sum_{i=1}^n \gamma_i C_{i,g}^{\mathcal{F}} + \epsilon_g^{\mathcal{F}} H(\gamma), \quad (1)$$

where $C_{i,g}^{\mathcal{F}} = \|\hat{z}_g^{\mathcal{F}} - z_{i,g}^{\mathcal{F}}\|_{M_g^{\mathcal{F}}}^{p_g^{\mathcal{F}}}$ is the cost (distance) between the incoming sample $\hat{z}$ and the i -th training sample at granularity g , with the metric tensor $\|\cdot\|_{M_g^{\mathcal{F}}}$ being calculated as:

$$\|u\|_{M_g^{\mathcal{F}}}^2 = u^T M_g^{\mathcal{F}} u, \quad (2)$$

where $M_g^{\mathcal{F}} \in \mathbb{R}^{d_g^{\mathcal{F}} \times d_g^{\mathcal{F}}}$ is learned from the training data to capture the relational structure of the features; $p_g^{\mathcal{F}} \geq 1$ is the order of the distance, typically set to $p_g^{\mathcal{F}} = 2$ for computational efficiency. The term $H(\gamma) = -\sum_{i=1}^n \gamma_i \log \gamma_i$ is the regularization, with $\gamma = (\gamma_1, \dots, \gamma_n) \in \Delta_n$ representing the transport plan, where $\Delta_n = \{\gamma \in \mathbb{R}^n : \sum_{i=1}^n \gamma_i = 1, \gamma_i \geq 0\}$ is the probability simplex, and $\epsilon_g^{\mathcal{F}} > 0$ controls the regularization strength.

The Gromov-Wasserstein distances [38] of temporal and spatial dimensions $\rho_{\text{GW},g}^{\mathcal{D}}(\hat{z}_g^{\mathcal{D}}, \mathcal{Z}_g^{\mathcal{D}})$, $\mathcal{D} \in \{\mathcal{T}, \mathcal{S}\}$ are computed as

$$\min_{\gamma \in \Delta_n} \sum_{i,j=1}^n \gamma_i \gamma_j \|\hat{z}_g^{\mathcal{D}} - z_{i,g}^{\mathcal{D}}\|_{M_g^{\mathcal{D}}}^2 \cdot \|\hat{z}_g^{\mathcal{D}} - z_{j,g}^{\mathcal{D}}\|_{M_g^{\mathcal{D}}}^2 + \epsilon_g^{\mathcal{D}} H(\gamma). \quad (3)$$

B. OOD Score Computation

The MDCG framework computes OOD scores by analyzing anomaly levels across all dimensions and granularity scales. For an incoming sample $\hat{z}$, the MDCG-specific OOD score corresponding to dimension $\mathcal{D}$ is computed as:

$$\psi^{\mathcal{D}}(\hat{z}) = \sum_{g \in \mathcal{G}^{\mathcal{D}}} \alpha_g^{\mathcal{D}} \cdot f_g^{\mathcal{D}}(\rho_g^{\mathcal{D}}(\hat{z}_g^{\mathcal{D}}, \mathcal{Z}_g^{\mathcal{D}})), \quad (4)$$

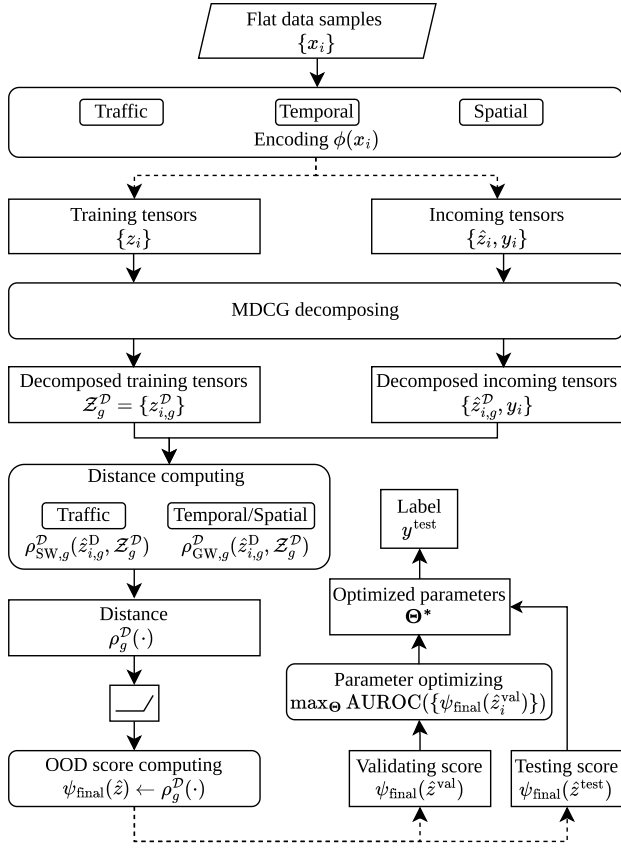


Fig. 4. Workflow of MDCG-IDS from training to inference. Flat data samples, including both training and incoming samples, are first encoded and represented as MDCG tensors. Then Wasserstein distances quantify the deviation of the incoming tensor from the benign distribution of training tensors, followed by OOD score calculation for the incoming tensors (i.e., validating and testing tensors which have labels y_i). Parameters are optimized using labeled validating tensors, after that the framework performs inference on the testing sample $\hat{z}^{\text{test}}$.

where $\alpha_g^D \geq 0$ represents the granularity-specific weight, $\rho_g^D(\cdot, \cdot)$ is SW or GW distance, and $f_g^D(\cdot)$ is the sigmoid function:

$$f_g^D(u) = \frac{1}{1 + \exp(-\lambda_g^D(u - \tau_g^D))}, \quad (5)$$

where $\lambda_g^D > 0$ controls the steepness, and $\tau_g^D \geq 0$ determines the midpoint of the sigmoid curve. The final OOD score is computed as:

$$\psi_{\text{final}}(\hat{z}) = \beta^F \cdot \psi^F(\hat{z}) + \beta^T \cdot \psi^T(\hat{z}) + \beta^S \cdot \psi^S(\hat{z}), \quad (6)$$

where $\{\beta^F, \beta^T, \beta^S\}$ are non-negative weights satisfying $\beta^F + \beta^T + \beta^S = 1$.

C. Optimization Framework

Let $\Theta = \{\{\alpha_g^D\}, \{\beta^F, \beta^T, \beta^S\}, \{\lambda_g^D, \tau_g^D\}, \{M_g^D, \epsilon_g^D, p_g^D\} : g \in \mathcal{G}^D, D \in \{\mathcal{F}, \mathcal{T}, \mathcal{S}\}\}$ represent the complete parameter set. The proposed optimization framework can be seen in Fig. 4, which summarizes the complete workflow of MDCG-IDS. In the figure, after the tensor encoding, distance measurement, and OOD score computation, an optimization framework is introduced to calculate optimized values of Θ with respect to

the validating tensors. The optimization objective maximizes the AUROC on the validating sample $\hat{z}_i^{\text{val}}$:

$$\Theta^* = \arg \max_{\Theta} \text{AUROC}(\{\psi_{\text{final}}(\hat{z}_i^{\text{val}})\}_{i=1}^{n^{\text{val}}}, \{y_i\}_{i=1}^{n^{\text{val}}}), \quad (7)$$

where $\{\psi_{\text{final}}(\hat{z}_i^{\text{val}})\}_{i=1}^{n^{\text{val}}}$ are the anomaly scores of the validating samples, n^{val} is the size of the validating set, and $\{y_i\}_{i=1}^{n^{\text{val}}}$ are the ground truth labels, with $y_i \in \{0, 1\}$ indicating normal (0) or anomalous (1) traffic. This optimization is subject to these constraints:

$$\sum_{g \in \mathcal{G}^D} \alpha_g^D = 1, \quad \forall D \in \{\mathcal{F}, \mathcal{T}, \mathcal{S}\}, \quad (8)$$

$$\alpha_g^D, \tau_g^D, p_g^D \geq 0, \quad \forall g \in \mathcal{G}^D, D \in \{\mathcal{F}, \mathcal{T}, \mathcal{S}\}, \quad (9)$$

$$\beta^F, \beta^T, \beta^S \geq 0, \quad (10)$$

$$\beta^F + \beta^T + \beta^S = 1, \quad (11)$$

$$\lambda_g^D, \epsilon_g^D > 0, \quad \forall g \in \mathcal{G}^D, D \in \{\mathcal{F}, \mathcal{T}, \mathcal{S}\}. \quad (12)$$

This optimization is performed offline, and during inference, the framework applies Θ^* directly for computing the score of the testing sample $\hat{z}^{\text{test}}$ (Fig. 4). The AUROC maximization objective provides a threshold-independent evaluation that accounts for both true positive and false positive rates.

D. Computational Complexity

The framework's computational cost comprises two phases: (1) MDCG tensor construction operates in $O(D \times G_{\text{max}} \times F_{\text{max}})$ per sample, which is linear in the input size; (2) OT distance computation using Sinkhorn-Knopp approximation scales as $O(n^2 \times k \times D \times G_{\text{max}})$, where n is the training set size, k is the number of Sinkhorn iterations, and $D \times G_{\text{max}}$ accounts for parallel computation across all dimension-granularity pairs. The Sinkhorn-Knopp approximation has reduced computational complexity of exact optimal transport from $O(n^3 \log n)$ [19] to $O(n^2 \times k)$. In practice n represents a manageable reference set (e.g., $10^3 - 10^4$ benign samples), and k typically ranges from 10 to 20.

V. EXPERIMENTS

A. Experiment Setup

1) *Datasets*: The proposed model is evaluated on four datasets: CIC-IDS-2017 (CIC17) [5], UNSW-NB15 (NB15) [39], TON-IoT (TON) [40], and 5GAD [41]. These datasets are chosen to cover both generic conventional enterprise networks (CIC17 and NB15) and specialized networks: IoT (TON) and 5G. Especially since 5G has specialized control-plane call flows generated sequentially, we can evaluate MDCG-IDS by capturing the temporal patterns of network traffic. This dataset contains specific attacks in 5G, such as *GetUserData*, *GetAllNFs*, and *FakeAMFInsert* (Table IV). The CIC17 dataset provides divert attack coverage, such as *DDoS*, several types of *DoS*, and *Web attacks*; while NB15 comprises 9 modern network attacks including *DoS*, *Reconnaissance*, and *Worms*. TON targets IoT environments with heterogeneous telemetry data from smart devices and sensors, containing attacks such as *dos*, *injection*, and *backdoor*.

We define two levels of OOD by selecting different attacks for hold-out, as described in Table IV. While the first level (OOD-1) simply holds out some attack classes, the second

TABLE IV

CLASS LABELS OF TRAINING, VALIDATING, AND TESTING SETS. TWO LEVELS OF OOD INSIDE THE TESTING SETS: OOD-1 SIMPLY HOLDS OUT SOME ATTACK CLASSES, WHILE OOD-2 HOLDS OUT ENTIRE ATTACK FAMILIES AND PROTOCOLS, AS WELL AS THE ATTACKS GENERATED BY A SPECIFIC TOOL

Dataset	Split	Type	Classes
CIC17	Training	Benign	<i>BENIGN</i>
	Validating	ID	<i>BENIGN, DoS Hulk, PortScan, DoS GoldenEye, FTP-Patator, SSH-Patator, DoS slowloris, DoS Slowhttptest</i>
	Testing	ID	All classes in validating set
		OOD-1	<i>DDoS, Web Attack – Brute Force, Web Attack – XSS</i>
		OOD-2	<i>Bot, Heartbleed, Infiltration, Web Attack – Sql Injection</i>
NB15	Training	Benign	<i>Normal</i>
	Validating	ID	<i>Normal, Fuzzers, DoS, Exploits, Generic, Reconnaissance</i>
	Testing	ID	All classes in validating set
		OOD-1	<i>Backdoor</i>
		OOD-2	<i>Shellcode, Worms, Analysis</i>
TON	Training	Benign	<i>normal</i>
	Validating	ID	<i>normal, scanning, dos, ddos, password</i>
	Testing	ID	All classes in validating set
		OOD-1	<i>xss, injection</i>
5GAD		OOD-2	<i>ransomware, mitm, backdoor</i>
	Training	Benign	<i>Normal</i>
	Validating	ID	<i>Normal, AMFLookingForUDM, GetAll-NFs, FakeAMFInsert, randomAMFInsert, CrashNRF, FakeAMFDelete</i>
	Testing	ID	All classes in validating set
		OOD-1	<i>GetUserData, randomDataDump</i>
		OOD-2	<i>automatedRedirectWithTimer, automatedDropWithTimer</i>

level (OOD-2) holds out entire attack families and protocols (e.g., the *Heartbleed* exploit in CIC17 is for OpenSSL) or attacks generated by specific tools. In this way, the testing samples will contain patterns that are less likely to be seen during training. Aligned with the attack categories in Fig. 1, an example of *ID-yet-unknown* attack in CIC17 could be a new variant of *DoS* that exhibits traffic patterns similar to the trained *DoS* classes (*DoS Hulk*, *DoS slowloris*, *DoS Slowhttptest*); while an *OOD-yet-known* attack could be *Heartbleed*, which is a known attack type in cybersecurity literature but was not covered by the training and validating sets of CIC17.

2) *Preprocessing*: To extract MDCG features from the datasets, we need their sample records at the flow and session level. The publicly available version of these labeled records mostly misses information on source and destination IPs (except for NB15). Therefore, we reconstruct their flow and session-level records from the raw capture version, to which we add IP addresses. The flow level is created based on the 5-tuples {source IP, destination IP, source port, destination port, protocol} with a 10-second inactive timeout and a 30-second active timeout. CIC17 labels are re-assigned using the published attack-period information, assisted by the CICFlowMeter tool [42]. For NB15 and TON, we re-label entries based on the released ground-truth tables, using the Payload-Byte tool [43] for NB15 and nProbe [44] for ToN. For 5GAD, we reconstruct flow-level records from raw captures that were already labeled by grouping. The feature *application protocol* in 5GAD is unavailable due to TLS encryption;

TABLE V

LIST OF MDCG FEATURES USED IN EXPERIMENTS

Dimension	Granularity	Features
Traffic (for each z)	Packet-level	Mean, standard deviation, min, and max of packet size, inter-arrival time, and TCP window size
	Flow-level	Per-flow total packets and bytes, flow duration, bytes per second
	Session-level	Variance of flow durations and inter-flow times, flow count, ratios of forward/backward packets, bytes
Temporal (for each z)	Subflow-level (0.5 s)	Symbolic sequence of TCP flags and transport protocols
	Flow-level (10 s)	Frequency and entropy of subflow-level TCP flag occurrences and transport protocol occurrences
	Session-level (60 s)	Variance and entropy of flow-level TCP flag counts and transport protocol counts
Spatial	Session-level (for each z)	Symbolic set of per-session TCP flags and transport protocols
	Host-level (for the IP pair of z)	Symbolic set of per-host destination ports and transport protocols

hence, this feature is excluded from all datasets (Table V) to maintain consistency.

The session level is built as bidirectional flows separated by TCP connection states (FIN/RST flags). For non-TCP traffic (UDP, ICMP), sessions are approximated using unidirectional flow segments grouped by common endpoints and protocol; these segments are aggregated if they are separated by no more than 60 seconds of inactivity. The experiments use default temporal granularity of three levels: subflow (0.5 s windows), flow (10 s windows), and session levels (60 s windows). These time windows are defined by referring to the standard flow-timeout conventions employed in NetFlow [45] and IPFIX implementations [46], [47], in which *active timeout* ranges from 120 seconds to 30 minutes and *idle timeout* is from 15 seconds to 5 minutes. After this reconstruction, the final datasets comprise 1.38 million sessions (CIC17), 1.04 million sessions (NB15), 1.18 million sessions (TON), and 23621 sessions (5GAD). Table V lists the features of all dimensions and granularities; these features are used by all baseline models.

3) *Model Implementation*: The proposed framework is developed in Python using PyTorch 2.1, with the POT library for optimal transport computations. Parameters including granularity weights $\{\alpha_g^D\}$, dimensional weights $\{\beta^F, \beta^T, \beta^S\}$, regularization $\{\epsilon_g^D\}$, and sigmoid parameters $\{\lambda_g^D, \tau_g^D\}$ are jointly optimized using gradient-based methods to maximize AUROC on the validating set. To deal with the non-differentiability in optimization exact form, a common approach is to use surrogate-based methods [48], [49]. Our non-differentiable AUROC is optimized through a surrogate computed from the validation scores $\{\psi_{\text{final}}(\hat{z}_i^{\text{val}})\}_{i=1}^{n^{\text{val}}}$ and labels $\{y_i\}_{i=1}^{n^{\text{val}}}$. Specifically, the objective can be viewed as maximizing a differentiable approximation of AUROC(Θ). Since all components of the MDCG scoring function $\psi_{\text{final}}(\hat{z})$ (i.e., including the SW distances $\rho_{\text{SW},g}^F$, the GW distances $\rho_{\text{GW},g}^D$, $D \in \{T, S\}$, the sigmoid mappings $f_g^D(\cdot)$, and the convex aggregation weights α_g^D, β^D) are differentiable with respect to Θ , we compute the gradients of AUROC(Θ) approximation

via backpropagation. Then optimization is performed using standard first-order gradient-based methods (Adam, learning rate 0.001) with early stopping based on validation AUROC.

We run experiments on an Intel i9 3.2 GHz CPU, 64 GB RAM, and an Nvidia LS40 GPU. Each dataset uses a split 60%/20%/20% where (1) the training set contains only benign traffic, and (2) the validating and testing sets contain stratified samples of both normal and anomalous traffic to maintain attack-type proportions. Hyperparameters are selected via 5-fold cross-validation within the validating set.

4) *Evaluation Metrics*: We evaluate the MDCG-IDS framework using both threshold-independent and threshold-dependent metrics. The primary metric is AUROC, which measures the model's ability to discriminate between normal and anomalous traffic across all possible decision thresholds, making it robust to the class imbalance issue. At the optimal operating points that are determined by Youden's index, we report the threshold-dependent metrics, including the False Positive Rate (FPR) and F1-score. The FPR is particularly critical because it quantifies benign traffic misclassified as attacks, which directly impacts operational feasibility. The experimental results reported in the following sections are computed over the entire testing set in Table IV (i.e., combining ID, OOD-1, and OOD-2 samples), with the exception of the attack-specific performance results which provide a per-category breakdown. All experimental results are reported as mean $\pm$ standard deviation across 5 independent runs with different random seeds.

5) *Baselines*: MDCG-IDS is compared against various baselines, including both closed-set K-class and open-set methods. For K-class baselines, models are trained on a standard K-class dataset and, at inference, assign every testing sample to one of the K classes, even if the sample comes from an unknown category. For open-set baselines, the models are designed to detect inputs that do not belong to the K known classes (i.e., rejecting them as unknown/OOD). We note that the K-class baselines are trained on the combination of the training and validating sets (Table IV).

a) *Closed-set k-class baselines*: We evaluate various closed-set baselines, including (i) traditional supervised learning methods: Support Vector Machine (SVM) and Random Forest, and (ii) state-of-the-art models: TCG-IDS [24], LS-GRU [22], and AHDM [26]. TCG-IDS is a temporal contrastive graph learning approach that constructs dynamic graphs from network flows and distinguishes attack patterns through temporal dependencies and graph structural information. We evaluate TCG-IDS with temporal window sizes of 30 s (TCG-IDS) and 60 s (TCG-IDS-60s) to assess the impact of temporal granularity on graph-based attack detection. The baseline model LS-GRU combines attention mechanisms with GRU for early network intrusion detection, enabling focused learning on critical features in traffic sequences. To evaluate the effects of attention mechanism in LS-GRU, we implement it with 64 and 128 hidden dimensions, named LS-GRU and LS-GRU-128, respectively. The AHDM approach is an autoencoder-based hybrid model specifically designed for small-sample intrusion detection. This model combines reconstruction-based anomaly detection with supervised classification components.

b) *Open-set baselines*: We evaluate the following traditional open-set baselines: Isolation Forest, which utilizes ensemble-based anomaly scoring through random data partitioning, and One-Class SVM (OSVM), which learns a decision boundary around benign samples for novelty detection. The state-of-the-art baselines include FOSS [30], Scalable-NIDS (S-NIDS) [31], and CVAE-EVT [32]. The model FOSS employs adaptive isolation mechanisms and multi-scale feature analysis to handle unknown attack patterns. We implement FOSS with a feature dimensionality of 64, while its variant, FOSS-128, is enhanced with a dimensionality of 128. We also evaluate FOSS-mg, a variant that equips MDCG feature processing to validate whether MDCG decomposition enhances isolation-based OOD detection. The baseline S-NIDS utilizes clustering and soft-boundary classification for hybrid novelty detection. S-NIDS is implemented with a feature dimensionality of 128, while S-NIDS-256 is enhanced with a dimensionality of 256. Similarly to FOSS-mg, we also implement the S-NIDS-mg variant for validating the MDCG decomposition. The baseline CVAE-EVT combines the Conditional Variational Auto-Encoder (VAE) with Extreme Value Theory for tail-based uncertainty modeling. We implement CVAE-EVT with 64 and 128 latent dimensions, named CVAE-EVT-64 and CVAE-EVT-128, respectively. CVAE-EVT-ot is the variant that is integrated with the optimal transport module to assess whether the Wasserstein distance can enhance VAE-based anomaly detection.

B. Experimental Results

1) *Separability of Feature Representation*: To evaluate the quality of MDCG features, Fig. 5 shows their t-SNE visualization on the CIC17 dataset (Fig. 5b) compared to flat features selected by mutual information (Fig. 5a). In Fig. 5a, we select 28 flat features that produce the highest mutual information between the NetFlow records and their class labels. To mitigate the class imbalance issue in CIC17, each class is capped at 10,000 samples; consequently, these t-SNE visualizations plot a total of 73,901 samples. Fig. 5a shows considerable overlaps among attack classes, particularly between OOD-2 attacks and ID classes. In contrast, Fig. 5b demonstrates substantially improved class separability, where individual attack types form more compact and distinguishable clusters. In particular, the benign and OOD classes exhibit a clear separation from other categories.

Fig. 5c illustrates the feature importance analysis that quantifies the contribution of each MDCG component to class separation. Traffic-level features (packet, flow, and session granularities) demonstrate high importance scores, while temporal and spatial dimensions provide complementary discriminative power. The cluster separation metrics (Fig. 5d) compare inter-cluster separation (i.e., the distance between different attack classes) and intra-cluster compactness (i.e., the tightness within each attack class) across three feature representations. The flatten features (28 features selected by mutual information) achieve a separation score and a compactness score of 0.342 and 0.289, respectively, while the single-dimensional representation (using only traffic dimension) shows moderate improvements to 0.456 and 0.378, respectively. MDCG representation achieves significantly higher

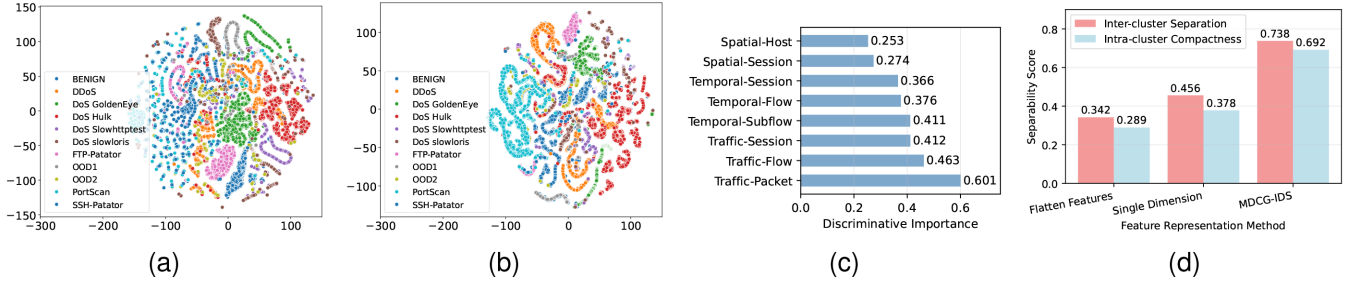


Fig. 5. t-SNE visualization of CIC17 using (a) Flat features selected by mutual information and (b) MDCG features. The MDCG representation is further analyzed on (c) Feature importance and (d) Separability score.

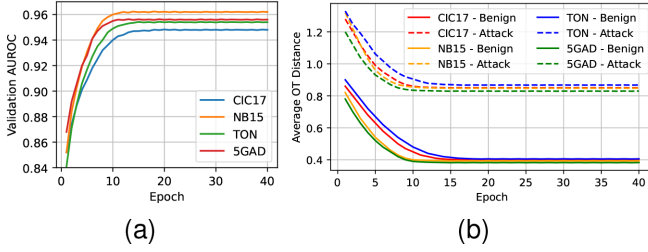


Fig. 6. Convergence and stability of optimization process over epochs across all datasets: (a) Average validation AUROC over 5 independent runs, and (b) Average OT distance over 5 independent runs for benign and attack samples.

scores (0.738 and 0.692), suggesting that the MDCG approach creates a more separable feature space.

2) *Optimization Convergence*: In Fig. 6, we illustrate the change of validation AUROC and OT distance during the optimization process. Fig. 6a shows that the validation AUROC increases rapidly in the early epochs and stabilizes after approximately 15–20 epochs across all datasets, with minor fluctuations. Consistently, Fig. 6b demonstrates that the average OT distances for benign and attack samples both decrease sharply in the initial phase and stabilize thereafter, with attack samples maintain a higher transport distance since they are structurally separated from benign embedding space. The convergence of AUROC and OT distance indicates that transport-based optimization helps the model learn stable and well-separated representations.

3) *Overall Performance Comparison*: To ensure a fair comparison, this experiment assesses all methods for binary intrusion detection instead of multi-class detection. In this way, each closed-set classifier outputs class predictions and then converts them to binary decisions, where predictions other than *Benign* are labeled as attacks. Unlike closed-set classifiers, open-set methods directly output anomaly scores for threshold-based binary classification.

Table VI presents the experimental results of all baselines across four datasets. MDCG-IDS consistently achieves superior performance in open-set detection, with AUROC scores of 0.948 (CIC17), 0.962 (NB15), 0.954 (TON), and 0.956 (5GAD). Our method outperforms all baselines, demonstrating a 3–7% improvement over the best competing methods. Among open-set baselines, CVAE-EVT shows competitive performance, while FOSS performs well on traditional datasets but exhibits degraded performance on IoT traffic (TON) and 5G traffic (5GAD). In terms of parameter-varying variants,

TABLE VI
PERFORMANCE COMPARISON OF MDCG-IDS FRAMEWORK AGAINST BASELINE METHODS ON: AUROC, F1-SCORE, AND FALSE POSITIVE RATE ACROSS FOUR DATASETS

Method	CIC17			NB15			TON			5GAD		
	AUROC	F1	FPR	AUROC	F1	FPR	AUROC	F1	FPR	AUROC	F1	FPR
SVM	0.750	0.681	0.356	0.743	0.668	0.387	0.721	0.634	0.421	0.729	0.652	0.398
Random Forest	0.781	0.715	0.321	0.768	0.701	0.342	0.745	0.670	0.378	0.754	0.689	0.363
TCG-IDS	0.834	0.768	0.245	0.820	0.756	0.267	0.798	0.731	0.298	0.803	0.739	0.284
TCG-IDS-60s	0.847	0.785	0.229	0.835	0.773	0.251	0.812	0.748	0.284	0.818	0.756	0.267
LS-GRU	0.806	0.741	0.283	0.792	0.728	0.301	0.769	0.690	0.334	0.774	0.712	0.318
LS-GRU-128	0.821	0.758	0.267	0.806	0.745	0.287	0.783	0.714	0.318	0.789	0.728	0.302
AHDM	0.803	0.738	0.289	0.789	0.723	0.308	0.766	0.692	0.341	0.771	0.706	0.325
Isolation Forest	0.831	0.769	0.276	0.823	0.756	0.290	0.809	0.731	0.321	0.795	0.723	0.307
OSVM	0.818	0.754	0.301	0.806	0.742	0.334	0.792	0.718	0.350	0.783	0.701	0.368
FOSS	0.897	0.864	0.150	0.891	0.842	0.182	0.856	0.801	0.237	0.864	0.819	0.205
FOSS-128	0.912	0.878	0.146	0.907	0.859	0.169	0.873	0.821	0.214	0.881	0.837	0.187
S-NIDS	0.909	0.871	0.145	0.902	0.853	0.167	0.878	0.823	0.201	0.876	0.831	0.192
S-NIDS-256	0.921	0.885	0.130	0.917	0.868	0.154	0.891	0.837	0.187	0.893	0.849	0.173
CVAE-EVT	0.931	0.895	0.121	0.924	0.881	0.134	0.896	0.847	0.178	0.907	0.867	0.152
MDCG-IDS-KL	0.923	0.881	0.124	0.911	0.876	0.138	0.916	0.871	0.131	0.907	0.867	0.142
MDCG-IDS-JS	0.934	0.892	0.115	0.926	0.889	0.127	0.931	0.885	0.119	0.923	0.881	0.128
MDCG-IDS	0.948	0.907	0.097	0.962	0.923	0.089	0.954	0.912	0.102	0.956	0.918	0.094

FOSS-128 shows modest improvements over base FOSS (1.6–2.0% average gain), and S-NIDS-256 also achieves limited gains (1.3–1.8% improvement) across datasets. To benchmark our OT-based distance, we compare its performance with MDCG-IDS-KL, which uses Kullback-Leibler (KL) divergence for distance measurement, and MDCG-IDS-JS, which utilizes Jensen-Shannon divergence. Table VI shows that MDCG-IDS-KL suffers significant performance degradation (0.907–0.923 AUROC), while MDCG-IDS-JS achieves slightly higher performance (0.923–0.934 AUROC) but still falls substantially short of MDCG-IDS.

4) *Attack-Specific Performance*: Fig. 7 illustrates the breakdown of performance across two levels of OOD defined in Table IV. The figure shows that MDCG-IDS consistently outperforms all baseline methods in both OOD-1 and OOD-2. Especially, the proposed model shows significant improvements for OOD-2 scenarios, such as with *Worms* in NB15, *MITM* in TON, and *automatedRedirectWithTimer* in 5GAD. Traditional open-set methods (i.e., Isolation Forest and OSVM) demonstrate low performance across all attack categories, particularly struggling with OOD-2 attacks. CVAE-EVT represents the strongest baseline, showing competitive performance in OOD-1 but still exhibits significant degradation in OOD-2 scenarios. The baseline variants reveal limited enhancements: FOSS-128 obtains an average improvement of 3.1% over base FOSS, while S-NIDS-256 achieves a gain of 3.2% over S-NIDS.

5) *Ablation Study Results*: Table VII presents the ablation studies that evaluate the contributions of the MDCG dimensions and their distance metric. Among the variants of MDCG-IDS, Equal Weighting is the variant that assigns uniform weights ($\alpha_g^D = 1/|\mathcal{G}^D|$, $\beta^D = 1/3$) instead of learning

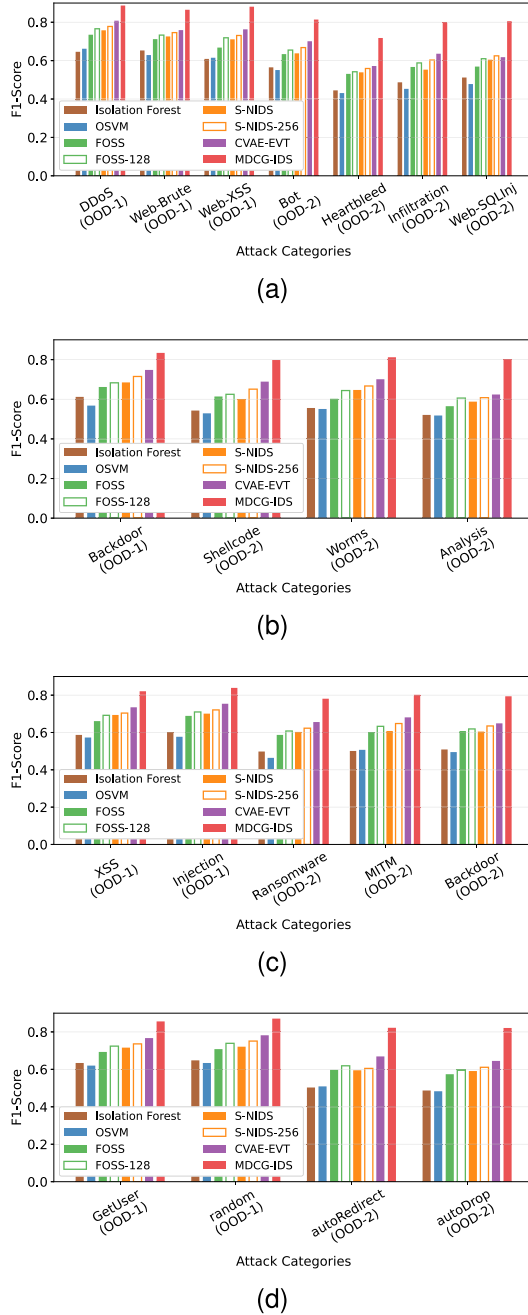


Fig. 7. Attack-specific performance comparison, showing F1-scores of OOD-1 and OOD-2 attacks for (a) CIC17, (b) NB15, (c) TON, and (d) 5GAD dataset.

them through parameter optimization. Feature Proxy is the MDCG-IDS version that approximates *application protocol* by *destination port*, while Full Feature uses complete *application protocol* when this information is available (i.e., in CIC17, NB15 and TON). The table shows that MDCG-IDS, Full Feature and Feature Proxy outperform their variants, including single- and pairwise-dimensional configurations (i.e., F-only, T-only, S-only, F+T, F+S, and T+S) and various other configurations (i.e., without MDCG, without OT, Euclidean Distance, and Equal Weighting). On 5GAD, the temporal dimension alone (that is, T-only) provides a strong result (0.928 AUROC), reflecting the importance of modeling the

TABLE VII
ABLATION STUDY RESULTS SHOWING THE IMPACT OF DIFFERENT DESIGN COMPONENTS OF MDCG-IDS. REPORTED VALUES ARE AUROC ON CIC17, NB15, TON, AND 5GAD DATASETS UNDER VARIOUS CONFIGURATIONS: SINGLE-DIMENSION ONLY, PAIRWISE DIMENSIONS, REMOVAL OF SPECIFIC MODULES, AND FEATURE APPROXIMATION

Configuration	CIC17 AUROC	NB15 AUROC	TON AUROC	5GAD AUROC
F-only	0.903	0.891	0.876	0.882
T-only	0.884	0.876	0.860	0.928
S-only	0.869	0.863	0.847	0.861
F+T	0.928	0.934	0.920	0.936
F+S	0.915	0.918	0.903	0.921
T+S	0.901	0.907	0.889	0.931
Without MDCG	0.892	0.879	0.865	0.874
Without OT	0.916	0.923	0.911	0.918
Euclidean Distance	0.930	0.934	0.926	0.932
Equal Weighting	0.925	0.941	0.932	0.939
Full Feature	0.949	0.962	0.954	-
Feature Proxy	0.948	0.962	0.954	0.952
MDCG-IDS	0.948	0.962	0.954	0.952

TABLE VIII
ABLATION STUDY SHOWING THE IMPACT OF DIFFERENT TEMPORAL WINDOW SIZES WHEN OTHER DIMENSIONS AND COMPONENTS OF MDCG-IDS ARE KEPT UNCHANGED

Window configurations	CIC17 AUROC	NB15 AUROC	TON AUROC	5GAD AUROC
0.5s/10s/60s	0.948	0.962	0.954	0.952
Only 0.5s	0.889	0.903	0.891	0.927
Only 10s	0.912	0.931	0.918	0.908
Only 60s	0.926	0.917	0.929	0.894
1s/20s/120s	0.941	0.951	0.946	0.935
0.25s/5s/30s	0.933	0.947	0.938	0.943

sequential control-plane call flows. Two-dimensional combinations show incremental improvements, with the F+T combination generally achieving the best pairwise performance across 4 datasets. Regarding the feature approximation, Table VII shows minimal impact on CIC17 when *application protocol* is unavailable (AUROC degraded from 0.949 to 0.948), while NB15 and TON show no degradation.

To assess the sensitivity to temporal window sizes, we evaluate the performances of alternative window configurations: short-only (0.5s), medium-only (10s), and long-only (60s), as well as alternative multi-scale combinations (1s/20s/120s and 0.25s/5s/30s) (Table VIII). This experiment shows the impacts of the alternative window configurations when all other components of MDCG-IDS are kept unchanged. Results of CIC17 show that single-window configurations achieve 0.889-0.926 AUROC, while multi-scale approaches consistently exceed 0.933 AUROC regardless of specific window values. The alternative combinations (1s/20s/120s: 0.941 AUROC; 0.25s/5s/30s: 0.933 AUROC) perform comparably to our default configuration (0.948 AUROC), with variations within 2%. Table VIII also reveals dataset-specific preferences for temporal scales: CIC17, NB15 and TON favor longer windows while 5GAD favors shorter windows; yet multi-scale configurations consistently achieve superior performance regardless of dataset type.

The cross-paradigm analysis (Table IX) presents the performance of baseline models with and without a MDCG module (-mg) and an optimal transport module (-ot). FOSS-mg

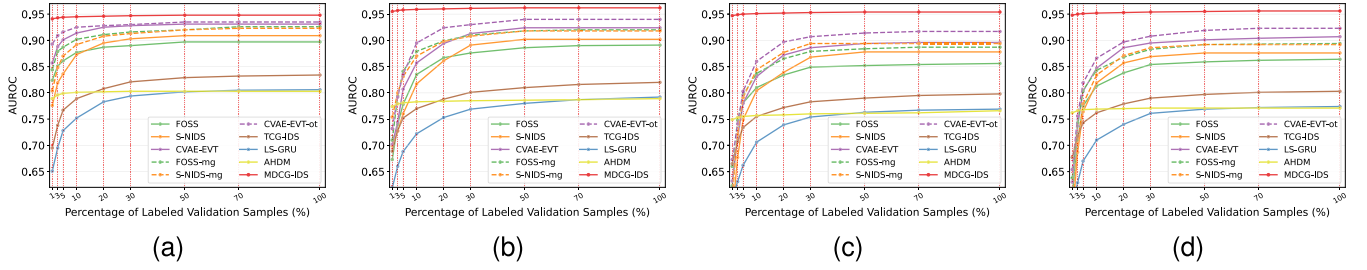


Fig. 8. AUROC performance with respect to the percentage of labeled validating data for (a) CIC17, (b) NB15, (c) TON, and (d) 5GAD datasets.

TABLE IX
AUROC COMPARISON OF BASELINE MODELS WITH AND WITHOUT MULTI-GRANULAR (–MG) AND OPTIMAL TRANSPORT (–OT) MODULE

Configuration	CIC17	NB15	TON	5GAD
FOSS/-mg	0.897/0.926	0.891/0.920	0.856/0.887	0.864/0.894
S-NIDS/-mg	0.909/0.923	0.902/0.918	0.878/0.894	0.876/0.892
CVAE-EVT/-ot	0.931/0.935	0.924/0.940	0.896/0.917	0.907/0.923

TABLE X
AUROC OF CROSS-DATASET EVALUATIONS IN WHICH THE MODELS ARE TRAINED ON ONE DATASET AND TESTED ON ANOTHER. THE LAST COLUMN IS THE AVERAGE DEGRADATION COMPARED TO THE INTRA-DATASET PERFORMANCE

Method	CIC17→NB15	NB15→CIC17	CIC17→5GAD	TON→5GAD	5GAD→TON	Avg. Degr.
Isolation Forest	0.651	0.645	0.598	0.672	0.681	19.8%
OSVM	0.628	0.621	0.576	0.651	0.659	21.2%
FOSS	0.768	0.760	0.716	0.798	0.812	14.3%
FOSS-128	0.794	0.786	0.739	0.821	0.834	12.8%
S-NIDS	0.788	0.781	0.723	0.806	0.819	13.9%
S-NIDS-256	0.805	0.798	0.746	0.829	0.841	12.1%
CVAE-EVT	0.808	0.801	0.751	0.834	0.846	11.8%
MDCG-IDS	0.894	0.887	0.859	0.923	0.937	6.2%

improves on average by 3.1% compared to FOSS, while S-NIDS-mg achieves a 1.6% improvement over S-NIDS. These results demonstrate the universal applicability and consistent effectiveness of MDCG processing. In the CVAE-EVT-ot variant, the Wasserstein distance is integrated as an additional reconstruction loss term that captures distributional differences in the VAE latent space. This enhanced variant gains an average improvement of 0.4% over CVAE-EVT.

6) *Cross-Dataset Evaluation*: We perform cross-dataset experiments in which one dataset is used for training and another for testing. Some dataset pairs are considered to represent more similar network domains (e.g., CIC17 and NB15), while other pairs represent less similar domains (e.g., CIC17 and 5GAD). The results in Table X show that the proposed framework demonstrates superior cross-dataset performance, with an average performance degradation of only 6.2%, compared to the 11.8–21.2% degradation observed in the baseline methods. Especially, MDCG-IDS achieves the highest performance, with scores of 0.923 and 0.937 in the scenarios TON → 5GAD and 5GAD → TON, respectively. These two datasets represent the IoT and 5G environments, which have similar device-heterogeneous and service-oriented characteristics. The CIC17 → 5GAD scenario represents the most

challenging transition: transferring from enterprise networks to 5G service-based environments, which leads to the lowest performance (0.859).

7) *Performance Over Amount of Validating Data*: Figure 8 illustrates the performance of all models against the percentage of the complete validating datasets. These validating data consist of 350,000 sessions for CIC17, 400,000 sessions each for NB15 and TON, and 23,621 sessions for 5GAD, which correspond to the minimum data required by LS-GRU (i.e., the model with the slowest convergence) to achieve its peak performance. The curves reveal distinct patterns among the approaches because they utilize different learning paradigms. Compared to its maximum performance, with only 3% labeled validating data, the proposed model reaches 99.47% on CIC17 (0.943 over 0.948 AUROC), 99.48% on NB15 (0.957 over 0.962 AUROC), 99.48% on TON (0.949 over 0.954 AUROC), and 99.37% on 5GAD (0.950 over 0.956 AUROC). MDCG-IDS stabilizes its performance beyond 20% labeled data. Especially, MDCG-IDS shows superiority over methods that are designed for small-sample data (i.e., AHDM, FOSS). AHDM achieves low initial performance and shows little improvement with increasing data. Supervised methods reveal a high dependency on labeled data, as TCG-IDS employs contrastive learning, and LS-GRU combines LSTM with GRU. These results demonstrate the effectiveness of parameter optimization through unsupervised optimal transport distances, requiring minimal labeled data solely for score weight calibration.

8) *Robustness Against Data Contamination*: Figure 9 illustrates the evaluations of the models under contaminated data, where attacks are mixed with benign training samples in a ratio of 0% to 15%. This scenario represents practical conditions in which obtaining perfectly clean data is challenging. These evaluations include both baseline models that use traditional unsupervised training (Isolation Forest, OSVM) and models that train mainly on benign traffic (FOSS, S-NIDS, and CVAE-EVT). The MDCG-IDS framework maintains a high detection accuracy against contamination, experiencing less than 6.7% degradation of AUROC at 15% contamination (on CIC17). Traditional unsupervised methods exhibit significant performance degradation, showing a decrease of more than 20% at 15% contamination. Other methods (FOSS, S-NIDS, CVAE-EVT) experience more than 14% degradation on all datasets. Enhanced variants (FOSS-mg, S-NIDS-mg, CVAE-EVT-ot) consistently outperform their base versions, but still significantly fall short of the MDCG-IDS framework.

9) *Computational Efficiency*: Fig. 10 presents the computational efficiency of the proposed framework compared

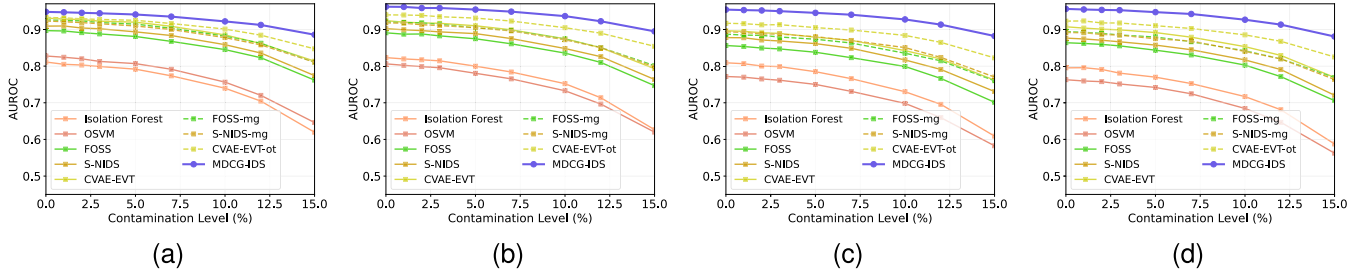


Fig. 9. Performance of the models against data contamination on (a) CIC17, (b) NB15, (c) TON, and (d) 5GAD. Attack samples are mixed into the benign training set at a ratio ranging from 0% to 15%.

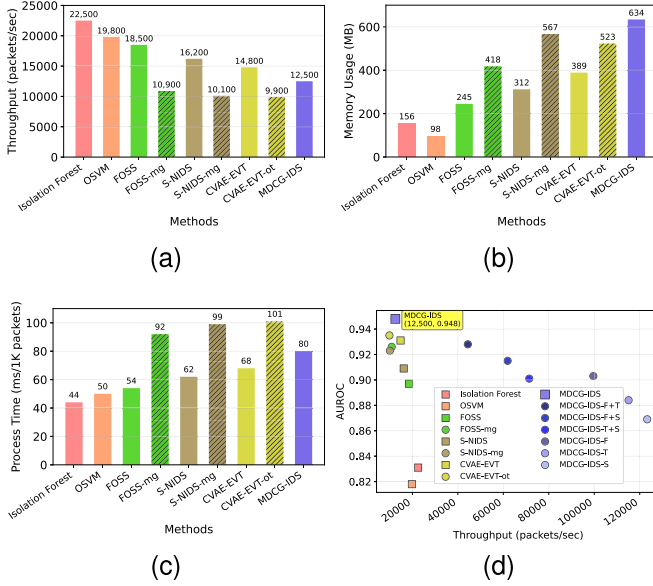


Fig. 10. Experimental results of the models about the computational efficiency on (a) Processing throughput in packets/second, (b) Memory consumption, (c) Processing time, and (d) Accuracy-efficiency trade-off.

to baseline methods. This experiment selects the CIC17 dataset as the representative benchmark due to its generic network environment and widespread adoption, which facilitates cross-study comparisons. Although our framework has a comprehensive multi-dimensional processing mechanism, it maintains competitive efficiency with a throughput of 12,500 packets per second, a memory consumption of 634 MB, and a processing time of 80 ms per 1000 packets. Within that multi-dimensional processing time, the OT computations which have $O(n^2 \times k \times D \times G_{\max})$ complexity contribute less than 15% of the total duration when the Sinkhorn approximation is utilized. Note that since we use network sessions as the data unit, throughput is reported in packets per second by tracking the number of raw packets contained within each session. Conventional open-set methods (Isolation Forest, OSVM) achieve the highest throughput of 19,800-22,500 packets per second with a minimal memory footprint (98-156 MB) but deliver poor detection performance (0.831 and 0.818 AUROC, respectively, as shown in Fig. 10d). The advanced baseline methods (FOSS, S-NIDS, CVAE-EVT) demonstrate a balanced performance-efficiency characteristic, with moderate throughput (14,800-18,500 packets per second) and reasonable memory consumption (245-389 MB). Their enhanced variants show considerable computational overhead: FOSS-mg takes

1.7 times longer, CVAE-EVT-ot takes 1.5 times longer, and S-NIDS-mg takes 1.6 times longer to process than their base versions. In Fig. 10d, the computational overhead of MDCG-IDS is justified by a substantial performance gain; especially, its variants form a clear Pareto front, illustrating the trade-off between detection performance and computational efficiency. Across the variants, when the dimension components are deactivated, throughput increases dramatically while AUROC decreases in a moderate manner.

VI. DISCUSSIONS AND FUTURE WORKS

A. Computational Trade-Offs

The proposed framework achieves improved performance with additional computational cost due to its comprehensive MDCG feature analysis. On the one hand, Fig. 10 shows that the framework achieves lower throughput than simpler baselines, but remaining within a comparable range to state-of-the-art methods (e.g., CVAE-EVT, S-NIDS), while delivering consistently higher detection accuracy. On the other hand, reflecting the throughput characteristics in other works such as [24] (standard enterprise network) and [50] (IoT network), our deployment targets edge servers or security gateways rather than high-speed backbone links, where comprehensive analysis is prioritized over raw line-rate processing. For resource-constrained environments, the framework targets aggregation points such as IoT gateway, rather than individual resource-limited endpoints. Future engineering optimizations will include self-optimizing sampling techniques, adaptive tensor-compute designs, and sparse tensor implementations.

B. Optimality

To mitigate the risk of local optima, we improve the convergence consistency by two common practices: multiple random initializations and early stopping [24], as described in Section V-A. Although global optimality cannot be guaranteed, (i.e., a known characteristic of gradient-based methods) the convergence in Fig. 6 as well as the low-variance curves in Fig. 8 and Fig. 9 demonstrate consistent performance across 5 independent runs with different initializations. Further, the evaluations throughout the paper, particularly the cross-dataset robustness analysis, show stable performance across four diverse datasets with varying traffic characteristics.

C. Deployment Architecture

Similar to the deployment mechanism in [51] which separates offline model update and online inference, the

MDCG-IDS's semi-supervised paradigm is designed for a hybrid deployment where the benign reference set and its corresponding parameters are updated offline, while real-time inference processes incoming traffic independently. In this way, the computationally intensive optimization is decoupled from the online detection which mainly relies on distance computation. Regarding the preparation of benign traffic, although this type of data already greatly reduces labeling effort, collecting purely benign records remains a common challenge for weakly-supervised, semi-supervised, cluster-based, and one-class detection frameworks [52]. Given this challenge, our framework demonstrates that it maintains robust performance (less than 6.7% AUROC degradation) even with 15% training set contamination (Section V-B8).

D. Feature Extraction

While most MDCG features are derived from packet headers and flow statistics without inspecting application-layer content, some features such as application protocol may be affected by encryption. In such cases, features in the MDCG tensor can be substituted with observable proxies (e.g., destination port as an approximation of application protocol under TLS) or zero-padded, as described in our 5GAD dataset processing. Such substitution or zero-padding strategies apply to other encryption mechanisms such as IPsec, which are challenges shared by NIDS frameworks.

E. Future Works

Several extensions would enhance the framework's practical deployability. First, given that the current temporal granularities are fixed based on flow timeout conventions, an adaptive granularity selection mechanism could automatically adjust granularity levels based on network characteristics. This adaptation can optimize the detection performance while reducing the computational overhead for different environments (e.g., bursty control-plane traffic vs. steady sensor streams). Second, the integration of federated learning would create opportunities for collaborative detection, wherein multiple organizations enhance their intrusion detection capabilities without sharing sensitive data. In this way, we address both privacy concerns and the challenge of limited diverse training data in individual organizations.

VII. CONCLUSION

In this paper, we developed a multi-dimensional cross-granularity framework for open-set network intrusion detection. The framework can decompose network samples into three complementary dimensions—traffic, temporal, and spatial—each at multiple granularity levels. The integration of optimal transport theory provides a novel foundation for quantifying the distributional deviations between benign and known/unknown network attacks. Our evaluations demonstrate substantial improvements in detection accuracy while addressing practical deployment challenges. Specifically, the framework consistently outperforms state-of-the-art methods across diverse experimental scenarios while requiring minimal labeled validating data. The reduced dependency on labeled traffic, combined with the training exclusively on benign samples, addresses the critical issue of data scarcity in practical

deployment. The practical applicability of the proposed framework is also validated by its reasonable processing throughput and resource consumption.

REFERENCES

- [1] A. Khraisat, I. Gondal, P. Vamplew, and J. Kamruzzaman, "Survey of intrusion detection systems: Techniques, datasets and challenges," *Cybersecurity*, vol. 2, no. 1, pp. 1–22, Dec. 2019.
- [2] N. Sultana, N. Chilamkurti, W. Peng, and R. Alhadad, "Survey on SDN based network intrusion detection system using machine learning approaches," *Peer-Peer Netw. Appl.*, vol. 12, no. 2, pp. 493–501, Mar. 2019.
- [3] (2023). *Malware Statistics*. [Online]. Available: <https://www.av-test.org/fileadmin/pdf/reports/AV-TESTHYASProtectEvaluationFeb.2023.pdf>
- [4] M. Ring, S. Wunderlich, D. Scheuring, D. Landes, and A. Hotho, "A survey of network-based intrusion detection data sets," *Comput. Secur.*, vol. 86, pp. 147–167, Sep. 2019.
- [5] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward generating a new intrusion detection dataset and intrusion traffic characterization," in *Proc. 4th Int. Conf. Inf. Syst. Secur. Privacy (ICISSP)*, 2018, pp. 108–116.
- [6] Z. Lin, Y. Shi, and Z. Xue, "IDSGAN: Generative adversarial networks for attack generation against intrusion detection," in *Proc. Pacific-Asia Conf. Knowl. Discovery Data Mining*, 2022, pp. 79–91.
- [7] W. J. Scheirer, A. de Rezende Rocha, A. Sapkota, and T. E. Boult, "Toward open set recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 7, pp. 1757–1772, Jul. 2013.
- [8] R. Sommer and V. Paxson, "Outside the closed world: On using machine learning for network intrusion detection," in *Proc. IEEE Symp. Secur. Privacy*, May 2010, pp. 305–316.
- [9] Y. Sharon, D. Berend, Y. Liu, A. Shabtai, and Y. Elovici, "TANTRA: Timing-based adversarial network traffic reshaping attack," *IEEE Trans. Inf. Forensics Security*, vol. 17, pp. 3225–3237, 2022.
- [10] P. García-Teodoro, J. Díaz-Verdejo, G. Maciá-Fernández, and E. Vázquez, "Anomaly-based network intrusion detection: Techniques, systems and challenges," *Comput. Secur.*, vol. 28, nos. 1–2, pp. 18–28, Feb. 2009.
- [11] X. Wang, X. Wang, M. He, M. Zhang, and Z. Lu, "Spatial-temporal graph model based on attention mechanism for anomalous IoT intrusion detection," *IEEE Trans. Ind. Informat.*, vol. 20, no. 3, pp. 3497–3509, Mar. 2024.
- [12] S. Cheng, D. Jin, Y. Ma, S. Chen, H. He, and J. Yang, "LinkGuard: Link flooding attack detection and mitigation via spatio-temporal graph convolutional network," *IEEE Trans. Netw. Sci. Eng.*, vol. 11, no. 5, pp. 4059–4075, Sep. 2024.
- [13] L. Chao, L. Yinghua, D. Fengqian, and S. Rui, "Spatio-temporal anomaly detection for 5G-clusters: A multi-scale fuzzy contrastive learning approach," *IEEE Trans. Netw.*, vol. 33, no. 4, pp. 1588–1602, Aug. 2025.
- [14] W. Yang, P. Gao, H. Huang, X. Wei, H. Zhang, and Z. Qu, "Advanced persistent threat detection in smart grid clouds using spatiotemporal context-aware graph embedding," in *Proc. IEEE Global Commun. Conf.*, Dec. 2022, pp. 534–540.
- [15] X. Jiang, H.-R. Zhang, and Y. Zhou, "Multi-granularity abnormal traffic detection based on multi-instance learning," *IEEE Trans. Netw. Service Manage.*, vol. 21, no. 2, pp. 1467–1477, Apr. 2024.
- [16] J. He, C. Tang, W. Li, T. Li, L. Chen, and X. Lan, "BR-HIDF: An anti-sparsity and effective host intrusion detection framework based on multi-granularity feature extraction," *IEEE Trans. Inf. Forensics Security*, vol. 19, pp. 485–499, 2023.
- [17] Y. Geng, Y. Li, and S. Zhang, "Research on multi-granularity intrusion detection algorithm based on sequential three-way decision," in *Proc. 5th Int. Conf. Electron. Inf. Technol. Comput. Eng.*, Oct. 2021, pp. 1154–1160, doi: [10.1145/3501409.3501613](https://doi.org/10.1145/3501409.3501613).
- [18] C. Villani, *Optimal Transport: Old and New*, vol. 338. Cham, Switzerland: Springer, 2009.
- [19] M. Cuturi, "Sinkhorn distances: Lightspeed computation of optimal transport," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2013, pp. 2292–2300.
- [20] R. Vinayakumar, M. Alazab, K. P. Soman, P. Poornachandran, A. Al-Nemrat, and S. Venkatraman, "Deep learning approach for intelligent intrusion detection system," *IEEE Access*, vol. 7, pp. 41525–41550, 2019.

- [21] C. Yin, Y. Zhu, J. Fei, and X. He, "A deep learning approach for intrusion detection using recurrent neural networks," *IEEE Access*, vol. 5, pp. 21954–21961, 2017.
- [22] T. E. T. Djaidja, B. Brik, S. Mohammed Senouci, A. Boulalouache, and Y. Ghamri-Doudane, "Early network intrusion detection enabled by attention mechanisms and RNNs," *IEEE Trans. Inf. Forensics Security*, vol. 19, pp. 7783–7793, 2024.
- [23] T. Bilot, N. E. Madhoun, K. A. Agha, and A. Zouaoui, "Graph neural networks for intrusion detection: A survey," *IEEE Access*, vol. 11, pp. 49114–49139, 2023.
- [24] C. Wu, J. Sun, J. Chen, M. Alazab, Y. Liu, and Y. Xiang, "TCG-IDS: Robust network intrusion detection via temporal contrastive graph learning," *IEEE Trans. Inf. Forensics Security*, vol. 20, pp. 1475–1486, 2025.
- [25] H. Zhang, K. Zeng, and S. Lin, "Federated graph neural network for fast anomaly detection in controller area networks," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 1566–1579, 2023.
- [26] N. Wei et al., "An autoencoder-based hybrid detection model for intrusion detection with small-sample problem," *IEEE Trans. Netw. Service Manage.*, vol. 21, no. 2, pp. 2402–2412, Apr. 2024.
- [27] J. Qin, Y. Xun, Z. Deng, and J. Liu, "GPIDS: GAN assisted contextual pattern-aware intrusion detection system for IVN," *IEEE Trans. Veh. Technol.*, vol. 73, no. 9, pp. 12682–12693, Sep. 2024.
- [28] B. Schölkopf, J. C. Platt, J. Shawe-Taylor, A. J. Smola, and R. C. Williamson, "Estimating the support of a high-dimensional distribution," *Neural Comput.*, vol. 13, no. 7, pp. 1443–1471, Jul. 2001.
- [29] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *Proc. 8th IEEE Int. Conf. Data Min.*, Dec. 2008, pp. 413–422. [Online]. Available: <https://api.semanticscholar.org/CorpusID:6505449>
- [30] Z. Zhao et al., "FOSS: Towards fine-grained unknown class detection against the open-set attack spectrum with variable legitimate traffic," *IEEE/ACM Trans. Netw.*, vol. 32, no. 5, pp. 3945–3960, Oct. 2024.
- [31] Z. Zhang, Y. Zhang, D. Guo, and M. Song, "A scalable network intrusion detection system towards detecting, discovering, and learning unknown attacks," *Int. J. Mach. Learn. Cybern.*, vol. 12, no. 6, pp. 1649–1665, Jun. 2021.
- [32] J. Yang, X. Chen, S. Chen, X. Jiang, and X. Tan, "Conditional variational auto-encoder and extreme value theory aided two-stage learning approach for intelligent fine-grained known/unknown intrusion detection," *IEEE Trans. Inf. Forensics Security*, vol. 16, pp. 3538–3553, 2021.
- [33] G. Ping and X. Ye, "Open-set intrusion detection with MinMax autoencoder and pseudo extreme value machine," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jul. 2022, pp. 1–8.
- [34] Y. Cao, K. Liu, Y. Lin, L. Wang, and Y. Xia, "Deep-reinforcement-learning-based self-evolving moving target defense approach against unknown attacks," *IEEE Internet Things J.*, vol. 11, no. 20, pp. 33027–33039, Oct. 2024.
- [35] Y. Liu, A. Zhao, L. Du, C. Zhang, H. Yan, and Z. Gu, "OIDMD: A novel open-set intrusion detection method based on Mahalanobis distance," in *Proc. 8th Int. Conf. Data Sci. CyberSpace (DSC)*, Aug. 2023, pp. 253–260.
- [36] J. Wu, H. Dai, K. B. Kent, J. Yen, C. Xu, and Y. Wang, "Open set dandelion network for IoT intrusion detection," *ACM Trans. Internet Technol.*, vol. 24, no. 1, pp. 1–26, Feb. 2024.
- [37] Y. A. Farrukh, S. Wali, I. Khan, and N. D. Bastian, "Detecting unknown attacks in IoT environments: An open set classifier for enhanced network intrusion detection," in *Proc. IEEE Mil. Commun. Conf. (MILCOM)*, Oct. 2023, pp. 121–126.
- [38] F. Mémoli, "Gromov–Wasserstein distances and the metric approach to object matching," *Found. Comput. Math.*, vol. 11, no. 4, pp. 417–487, Aug. 2011.
- [39] N. Moustafa and J. Slay, "UNSW-NB15: A comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set)," in *Proc. Mil. Commun. Inf. Syst. Conf. (MilCIS)*, Nov. 2015, pp. 1–6.
- [40] A. Alsaedi, N. Moustafa, Z. Tari, A. Mahmood, and A. Anwar, "TON IoT telemetry dataset: A new generation dataset of IoT and IIoT for data-driven intrusion detection systems," *IEEE Access*, vol. 8, pp. 165130–165150, 2020.
- [41] C. Coldwell et al., "Machine learning 5G attack detection in programmable logic," in *Proc. IEEE Globecom Workshops (GC Wkshps)*, Dec. 2022, pp. 1365–1370.
- [42] A. H. Lashkari, G. D. Gil, M. S. I. Mamun, and A. A. Ghorbani, "Characterization of Tor traffic using time based features," in *Proc. 3rd Int. Conf. Inf. Syst. Secur. Privacy (ICISSP)*, Porto, Portugal, Feb. 2017, pp. 253–262.
- [43] Y. A. Farrukh, I. Khan, S. Wali, D. Bierbrauer, J. A. Pavlik, and N. D. Bastian, "Payload-byte: A tool for extracting and labeling packet capture files of modern network intrusion detection datasets," in *Proc. IEEE/ACM Int. Conf. Big Data Comput., Appl. Technol. (BDCAT)*, Dec. 2022, pp. 58–67.
- [44] *NProbe*. Accessed: Apr. 30, 2025. [Online]. Available: <https://www.ntop.org/products/netflow/nprobe/>
- [45] B. Claise, "Cisco systems netflow services export version 9," Internet Eng. Task Force, RFC Editor, Tech. Rep. RFC 3954, Oct. 2004, doi: [10.17487/RFC3954](https://doi.org/10.17487/RFC3954).
- [46] R. Hofstede et al., "Flow monitoring explained: From packet capture to data analysis with NetFlow and IPFIX," *IEEE Commun. Surveys Tuts.*, vol. 16, no. 4, pp. 2037–2064, 4th Quart., 2014.
- [47] G. Sadasivan, N. Brownlee, B. Claise, and J. Quittek, *Architecture for IP Flow Information Export*, document RFC 5470, Mar. 2009. [Online]. Available: <https://www.rfc-editor.org/rfc/rfc5470.txt>
- [48] L. Yang, A. Moubayed, and A. Shami, "MTH-IDS: A multitiered hybrid intrusion detection system for Internet of Vehicles," *IEEE Internet Things J.*, vol. 9, no. 1, pp. 616–632, Jan. 2022.
- [49] P. Zhao, Y.-J. Zhang, and Z.-H. Zhou, "Exploratory machine learning with unknown unknowns," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, no. 12, pp. 10999–11006.
- [50] N. Chaabouni, M. Mosbah, A. Zemmari, C. Sauvignac, and P. Faruki, "Network intrusion detection for IoT security based on learning techniques," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 3, pp. 2671–2701, 3rd Quart., 2019.
- [51] C. Wu et al., "WAFBooster: Automatic boosting of WAF security against mutated malicious payloads," *IEEE Trans. Dependable Secure Comput.*, vol. 22, no. 2, pp. 1118–1131, Mar. 2025.
- [52] G. Pang, C. Shen, L. Cao, and A. V. D. Hengel, "Deep learning for anomaly detection: A review," *ACM Comput. Surv. (CSUR)*, vol. 54, no. 2, pp. 1–38, 2021.



Minh-Thuyen Thi received the Ph.D. degree in information and communications from Inje University, South Korea, in 2017. Since 2018, he has been a Post-Doctoral Researcher with Université Paul Sabatier, France. Since 2019, he has been a Researcher with the Communications Systems Laboratory, Institute LIST, CEA, Université Paris-Saclay. Since 2024, he has been a Research Fellow with the National University of Singapore, Singapore. His research interests include communication networks, game theory-based network control, AI-based network management, and machine learning for network intrusion detection.



Mohan Gurusamy (Senior Member, IEEE) received the Ph.D. degree in computer science and engineering from Indian Institute of Technology Madras in 2000. He joined the National University of Singapore (NUS) in June 2000, where he is currently an Associate Professor with the Department of Electrical and Computer Engineering (ECE). He has over 280 publications to his credit, including two books and three book chapters. His research experience and interests include the areas of the Internet of Things, 5G, software defined networks, metaverse, edge computing, and optical networks. He served as the TPC Co-Chair for several conferences, including IEEE GLOBECOM 2019 (ONS) and IEEE ICC 2008 (ONS). He served as an Editor for IEEE TRANSACTIONS ON CLOUD COMPUTING, IEEE INTERNET OF THINGS JOURNAL, and Photonic Network Communications; and a Guest Editor for IEEE Communications Magazine and Computer Networks journal. He is serving on the Editorial Board for Computer Networks journal (Elsevier).